

1. Aim

To design and implement a web-based **Student Scholarship Eligibility Check** application using HTML and JavaScript. The objective is to validate user-inputted student marks, enforce boundary limits (0 to 100), evaluate tiers of scholarship rewards using conditional control structures, and dynamically update the document object model (DOM) to display results.

2. Theory

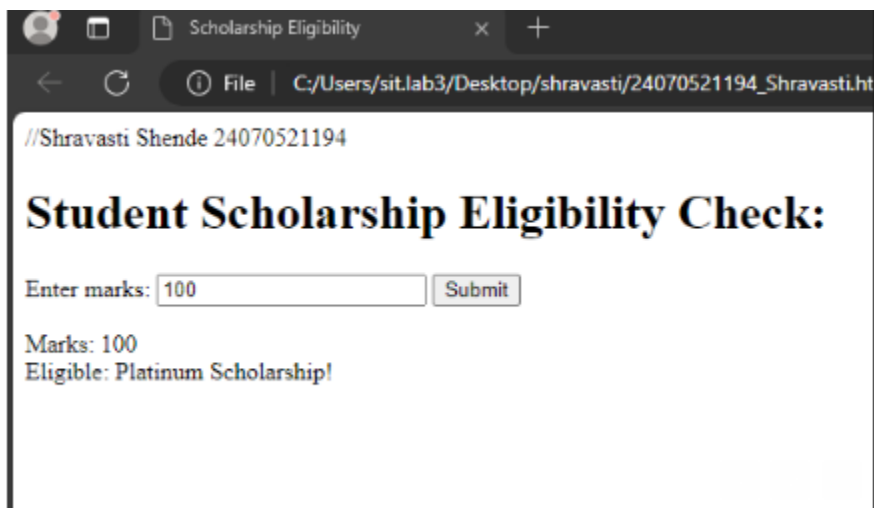
- **HTML Client Layout:** Uses structural elements such as headers (`<h1>`), input controls (`<input type="number">`), and action triggers (`<button>`) to construct an accessible user interface for collecting marks from the evaluator.
 - **Data Validation and Boundary Safeguards:** The program uses a strict logical layout to intercept and address illegal runtime inputs:
 - *Empty Field Check:* Ensures an entry is present; otherwise, it stops execution using a `return` statement and displays a browser alert.
 - *Range Validation:* Uses logical OR operators (`||`) to check if inputted marks fall outside the legitimate spectrum (`<0` or `>100`), avoiding illogical evaluation data.
 - **Conditional Control Tiers (If-Else If Ladder):** The application evaluates the input across multiple decision paths to determine specific tiers:
 - **≥ 95:** Platinum Scholarship
 - **≥ 85 and < 95:** Gold Scholarship
 - **≥ 70 and < 85:** Silver Scholarship
 - **≥ 50 and < 70:** Bronze Scholarship
 - **< 50:** Not Eligible
 - **Dynamic DOM Manipulation:** Leverages JavaScript's native document object referencing system (`document.getElementById()`). It extracts numerical raw input through the `.value` parameter and renders the evaluated results on the webpage by writing strings into the `.innerHTML` property of a target paragraph element.
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3. Code

```
<!DOCTYPE html>          //Shravasti Shende 24070521194
<html>
  <head>
    <title> Scholarship Eligibility </title>
  </head>
  <body>
    <h1> Student Scholarship Eligibility Check:</h1>
    Enter marks:
    <input type = "number" id = "marks">
    <button onclick = "checkEligibility()">Submit</button>
    <p id = "result"></p>
    <script>
      function checkEligibility()
      {
        var marks = document.getElementById("marks").value;
        if (marks == "")
        {
          alert("Please Enter Marks: ");
          return;
        }
        if (marks< 0 || marks> 100)
        {
          alert("Enter marks between 0 and 100");
          return;
        }
        if (marks>=95)
          sch = "Platinum Scholarship!";
        else if (marks >= 85)
          sch = "Gold Scholarship!";
        else if (marks >= 70)
          sch = "Silver Scholarship!";
        else if (marks>=50)
          sch = "Bronze Scholarship!";
        else
          sch = "Not Eligible";
        document.getElementById("result").innerHTML = "Marks: " + marks +
        "<br> Eligible: " + sch;
```

```
    }  
  </script>  
</body>  
  
</html>
```

4 output :



5 Conclusion

In this experiment, a functional **Student Scholarship Eligibility Check** utility was successfully created under the user identity of **Shravasti Shende (24070521194)**. The implementation shows how client-side JavaScript can validate numeric ranges and execute conditional logic.

By applying an `if-else if` control layout alongside input verification filters, the web system filters out invalid marks while correctly organizing passing scores into their respective scholarship brackets (Platinum, Gold, Silver, or Bronze) without requiring server-side processing.
